WANG XIN

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Education

2018.09 — Pres. **M.Eng.**, Artificial Intelligence Education Division, Central China Norma University (CCNU)
Supervisor: Qiusha Min

2016.02 — 2016.07 **Exchange Student**, College of Science, Providence University (PU)

2014.09 — 2018.07 **B.Eng.**, School of Biological and Chemical Engineering, Nanyang Institute of Technology (NYIST)

Research Interests

- My researches focus on machine learning and image processing;
- My research interest lies at the intersection of machine learning, deep learning, computer vision, behavior recognition, image compression, image segmentation, and 2D image processing and 3D visualization.

Research Experience

Artificial Intelligence Education Division, Central China Normal University
Master student **2018.09 — Pres.**

- **Deep Learning and Image Compressions**: The recent research is a novel compression technique that combines anatomical information and deep neural network [2,3]. According to the characteristics of medical image datasets, the medical image data set is divided into 12 organ areas by image segmentation, and each area is processed by discriminative model (fall, 2019).
- **Image Processing**: Different technologies (Oracle Java, Adobe Flash, and HTML5) are used to develop a web-based medical imaging application [1]. Simultaneously, this application connects to a medical image server and can provides several required functions for radiological interpretation, e.g. navigation, magnification, windowing, maximum intensity projection, fly-through, surface rendering, and volume rendering. Java-, Flash-, and HTML5-based medical imaging applications were tested on different operating systems over a local area network and a wide area network. (fall, 2018).
- **Machine Learning and Deep Learning**: In addition to the machine learning courses in the teaching plan, I also learned machine learning and deep learning on the Coursera online platform (fall, 2018).
  - Machine Learning on the Coursera platform provided by Stanford University
    Lectures: Andrew Ng
  - Deep Learning on the Coursera platform provided by Stanford University
    Lectures: Andrew Ng, Teaching Assistant-Kian Katanforoosh & Teaching Assistant-Younes Bensouda Mourri
    (1) Neural networks and deep learning
    (2) Improving deep neural networks: hyperparameter tuning, regularization and optimization
    (3) Structuring machine learning projects
    (4) Convolutional neural networks
    (5) Sequence models

Teaching Experiences

Course Coordinator, September 2018 – January 2019.
Central China Normal University Wollongong Joint Institute, Reasoning and Learning.

- Led the lab session for 90 students in Multimedia Technologies class. Fall, 2018

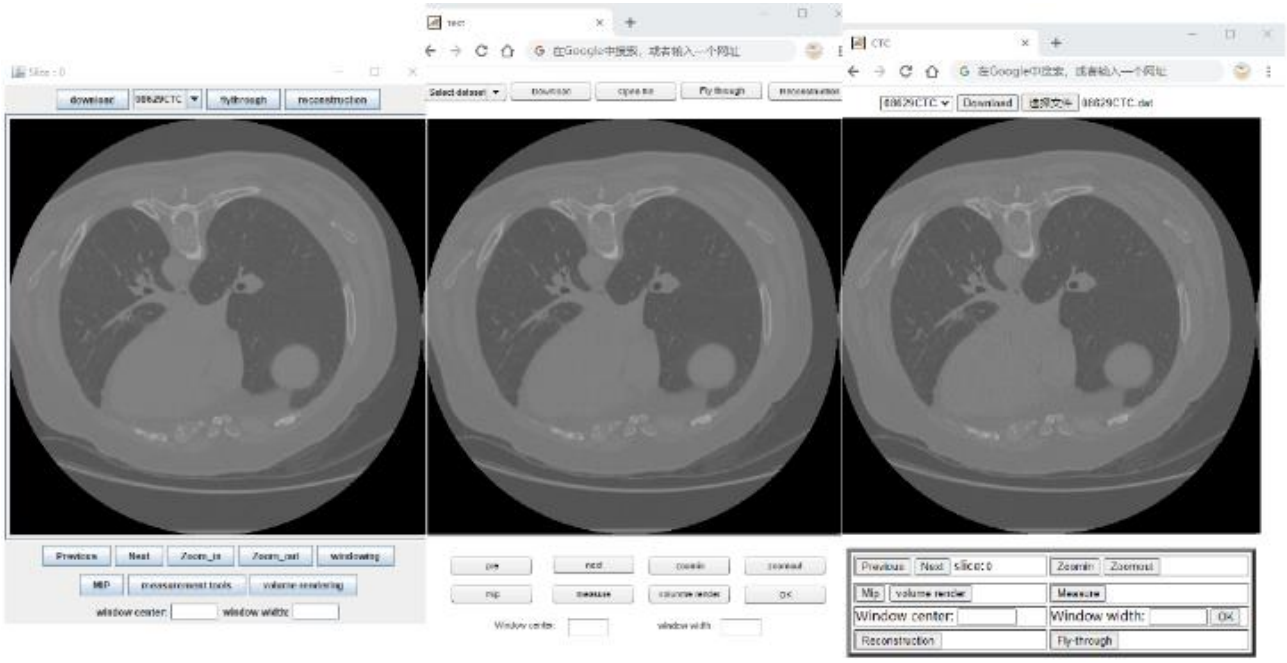
Teaching Assistant, September 2019 – January 2020.
Central China Normal University Wollongong Joint Institute, Reasoning and Learning.

- Teaching assistant for the graduate-level course CSCI924_JI119 – Reasoning and Learning, which is oriented to master students in the Central China Normal University Wollongong Joint Institute, Central China Normal University. Fall, 2019

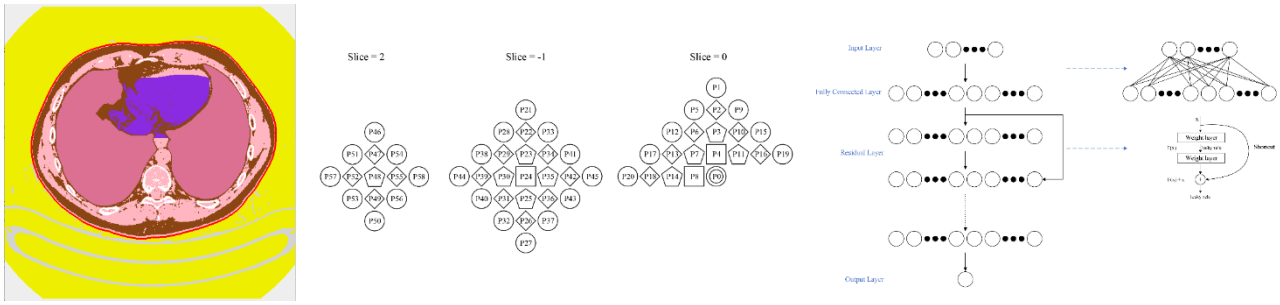
Publications

[1] Min Q*, Wang X*, Huang B*, et al. Web-Based Technology for Remote Viewing of Radiological Images: App Validation[J]. Journal of Medical Internet Research, 2020, 22(9): e16224. (SCI)

*these authors contributed equally



[2] Qiusha Min, Xin Wang. Lossless medical image compression based on region segmentation and residual neural network, 2020. (In Preparation)



[3] Qiusha Min, Neng Liu, Xin Wang. Lossless medical image compression based on region segmentation and decision tree, 2020. (In Preparation)

Patents & Software Copyright

[4] Remote medical image access system based on Java technology, Qiusha Min, Xin Wang, Software copyright, 2019SR0982502
[5] Remote medical image access system based on HTML5 technology, Qiusha Min, Xin Wang, Software copyright, 2019SR0982496
[6] Remote medical image access system based on Flash technology, Qiusha Min, Xin Wang, Software copyright, 2019SR0982505

Skills

- Computing Skills: Python, MATLAB, Java, TensorFlow, Keras.
- Language: Chinese (native), English (fluent).
- Object oriented programming-Java (on the Massive Open Online Courses platform provided by Zhejiang University).